

RESEARCH ARTICLE



Evaluating the effectiveness of artificial intelligence-based tools in detecting and understanding sleep health misinformation: Comparative analysis using Google Bard and OpenAI ChatGPT-4

Sergio Garbarino^{1,2} | Nicola Luigi Bragazzi^{1,3,4}

¹Department of Neuroscience, Rehabilitation, Ophthalmology, Genetics and Maternal, Child Sciences (DINOEMI), University of Genoa, Genoa, Italy

²Post-Graduate School of Occupational Health, Università Cattolica del Sacro Cuore, Rome, Italy

³Laboratory for Industrial and Applied Mathematics (LIAM), Department of Mathematics and Statistics, York University, Toronto, Ontario, Canada

⁴Human Nutrition Unit (HNU), Department of Food and Drugs, University of Parma, Parma, Italy

Correspondence

Nicola Luigi Bragazzi, Human Nutrition Unit (HNU), Department of Food and Drugs, Medical School, Building C, Via Volturno, 39, 43125 Parma, Italy.
Email: nicolaluigi.bragazzi@unipr.it

Summary

This study evaluates the performance of two major artificial intelligence-based tools (ChatGPT-4 and Google Bard) in debunking sleep-related myths. More in detail, the present research assessed 20 sleep misconceptions using a 5-point Likert scale for falseness and public health significance, comparing responses of artificial intelligence tools with expert opinions. The results indicated that Google Bard correctly identified 19 out of 20 statements as false (95.0% accuracy), not differing from ChatGPT-4 (85.0% accuracy, Fisher's exact test $p = 0.615$). Google Bard's ratings of the falseness of the sleep misconceptions averaged 4.25 ± 0.70 , showing a moderately negative skewness (-0.42) and kurtosis (-0.83), and suggesting a distribution with fewer extreme values compared with ChatGPT-4. In assessing public health significance, Google Bard's mean score was 2.4 ± 0.80 , with skewness and kurtosis of 0.36 and -0.07 , respectively, indicating a more normal distribution compared with ChatGPT-4. The inter-rater agreement between Google Bard and sleep experts had an intra-class correlation coefficient of 0.58 for falseness and 0.69 for public health significance, showing moderate alignment ($p = 0.065$ and $p = 0.014$, respectively). Text-mining analysis revealed Google Bard's focus on practical advice, while ChatGPT-4 concentrated on theoretical aspects of sleep. The readability analysis suggested Google Bard's responses were more accessible, aligning with 8th-grade level material, versus ChatGPT-4's 12th-grade level complexity. The study demonstrates the potential of artificial intelligence in public health education, especially in sleep health, and underscores the importance of accurate, reliable artificial intelligence-generated information, calling for further collaboration between artificial intelligence developers, sleep health professionals and educators to enhance the effectiveness of sleep health promotion.

KEYWORDS

generative conversational artificial intelligence, public health education, reliability analysis, sleep disorders, sleep health, sleep-related false myths, text-mining

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1 | INTRODUCTION

Good sleep is essential for the health and well-being of individuals of all ages, significantly impacting cognitive functions, mood, mental health and various aspects of physical health, including immunological, cardiovascular, cerebrovascular and metabolic conditions (Ramar et al., 2021). Proper sleep helps prevent accidents and injuries due to fatigue, such as those occurring at the workplace or while driving (Magnavita & Garbarino, 2017). Both short-term sleep deprivation and chronic sleep issues can have detrimental effects on physical and mental health-related quality of life, and public safety. Chronic lack of sleep is linked to a higher mortality risk, and exacerbates public health challenges related to both communicable and non-communicable diseases, such as cardiovascular disorders, diabetes, obesity and cancer (Magnavita & Garbarino, 2017; Ramar et al., 2021).

As such, promoting healthy sleep and addressing sleep disorders have become imperative (Lim et al., 2023). These efforts can improve both individual and societal well-being and productivity. Moreover, they emphasize the increasing importance and role of digital tools like websites, sleep trackers and apps in sleep health education within contemporary society (Tsai et al., 2022).

Nowadays, the Internet is a useful platform for health-related information, including sleep-related information, offering accessibility and cost-effectiveness, as evidenced by data indicating that more than half of American and European households utilized the Internet for health-related activities in 2019–2020, according to the US “National Telecommunications and Information Administration” (NTIA) and Eurostat, respectively (Cao & Goldberg, 2020; Eurostat, 2021). These activities included communicating with health professionals, accessing health records and seeking health information, such as websites and other digital resources related to sleep and sleep issues (McKenzie, 2024). The Internet is, indeed, a vital venue that provides the latest sleep research and personal guidance, with growing popularity in telemedicine and online expert consultations. Digital tools enable sleep pattern monitoring, promoting self-awareness and sleep quality improvement (McKenzie, 2024; Tsai et al., 2022).

In the artificial intelligence (AI) era, platforms like ChatGPT afford new opportunities for sleep health promotion (Perez-Pozuelo et al., 2020). ChatGPT and similar AI-based agents can provide instant, tailored sleep health advice, sleep hygiene tips and even relaxation techniques customized to individual needs. Moreover, as previously mentioned, the pervasive nature of AI in everyday technology allows for continuous monitoring and analysis of sleep patterns, offering insights that can lead to better sleep habits and overall health. Therefore, the future of sleep health is being shaped by a “data-driven revolution” in sleep science and medicine, leveraging advancements in AI, machine learning (ML) and wearable technologies. This revolution may contribute to the understanding, diagnosis and treatment of sleep disorders, as well as to customizing sleep health management (Bragazzi et al., 2019; Perez-Pozuelo et al., 2020; Redline & Purcell, 2021). However, accuracy and reliability are crucial, as AI can also exhibit biases (Gichoya et al., 2023; Seyyed-Kalantari et al., 2021) and spread

misinformation, which is a significant concern, particularly as AI technologies become more sophisticated. AI can, indeed, generate realistic text, audio and video content, which can be used maliciously to create and disseminate false information, deepfakes and other deceptive content. All this holds serious implications for individuals and societies (Zhou et al., 2023).

For instance, ChatGPT has been found to produce incorrect text and even fake scholarly citations. These “hallucinations” (Athaluri et al., 2023; Goddard, 2023), “confabulations” (McGowan et al., 2023), “fabrications” (Emsley, 2023) or “falsifications” (Emsley, 2023) have “been linked to a disturbance in language production, with probabilistic outputs based on estimates of semantic similarity”, allowing “informed guesses, with bits of false information being mixed with factual information” (Emsley, 2023).

Given these concerns, recently, we (Bragazzi & Garbarino, 2024) examined ChatGPT-4’s accuracy in addressing sleep health and debunking sleep myths (Robbins et al., 2019), informing efforts to promote sleep health. ChatGPT-4 was tasked with assessing 20 sleep-related misconceptions, identified by a panel of 10 experts, using a 5-point Likert scale. The study also measured sensitivity, positive predictive value and inter-rater agreement by means of the intra-class correlation (ICC) coefficient. ChatGPT-4 classified 85% of the statements as “false” (45%) or “generally false” (40%). It performed well in categories like “sleep timing”, “sleep duration” and “behaviors during sleep”, but varied in others like “pre-sleep behaviours” and “brain function and sleep”. On the Likert scale, ChatGPT-4 scored an average of 3.45 for falseness and 3.15 for public health significance, indicating good accuracy and understanding of the myths’ impact. Sensitivity was 85%, with a perfect positive predictive value of 100%, suggesting high reliability when labelling statements as false. Comparisons with expert ratings showed high ICC coefficients (0.83 for falseness and 0.79 for public health significance), indicating good alignment with expert opinions.

However, ChatGPT represents a specific instantiation of a broader class of AI known as “large language models” (LLMs), which are designed to understand human language and generate content that mimics it (Kim et al., 2024). All these models employ similar foundational AI principles, including transformer technology—a self-attention mechanism that allows the capturing and weighing of dependencies and contextual relations among different parts of the input text (known as “tokens”). In other words, this mechanism is akin to focusing more on relevant information while filtering and sidelining less pertinent details. However, if these models share the same technology, they exhibit different transformer architectures, designs and functionalities. While LLMs like ChatGPT have a model architecture based on a single-layer decoder-only transformer, LLMs like Google Bard employ a dual structure consisting of both an encoder (which processes and understands the input text) and a decoder (which generates the output). The latter structure potentially allows for a more nuanced understanding and production of text, enhancing the model’s capability to handle a wide array of language processing tasks. Moreover, the two AI-based chatbots differ in terms of training data (both quantitatively and qualitatively, the amount and the types of *corpora*

that have been utilized to calibrate and train the models), optimization techniques and fine-tuning strategies (which involve refining and making adjustments to a pre-trained model to adapt it to specific tasks or improve its performance on particular types of data), among others.

Therefore, differences between the two major classes of AI conversational agents in providing responses to medical issues can be expected. However, while the various versions of ChatGPT (ChatGPT-3.5 and ChatGPT-4) are knowledgeable in sleep disorders (Bilal et al., 2023; Campbell et al., 2023; Mira et al., 2023), even if to varying degrees, and recently passed sleep medicine exams (Cheong et al., 2023), Bard's capabilities in this area are yet to be explored, especially using a systematic and comparative approach (Bragazzi & Garbarino, 2024).

As such, besides ChatGPT, the present study also introduced Google Bard, with the primary aim of not only highlighting their similarities and differences in providing sleep-related information but also emphasizing their roles in public health and sleep health promotion. This comparison is vital for understanding how each AI-based tool can address misconceptions and contribute to the broader goals of enhancing community awareness of the importance of sleep hygiene practices and improving overall sleep health.

2 | MATERIALS AND METHODS

2.1 | List of 20 sleep expert-identified “false myths” on sleep

A collection of 20 sleep-related myths was identified by scanning the literature. A study (Robbins et al., 2019) was found, in which the authors compiled a list of common misconceptions about sleep obtained through a variety of sources—online searches of both popular media and scientific literature—and leveraged a Delphi method with input from a panel of 10 experts in sleep medicine and research. These myths were organized into six categories: (i) sleep duration (6 myths); (ii) sleep timing (1 myth); (iii) behaviours during sleep (4 myths); (iv) daytime behaviours affecting sleep (2 myths); (v) pre-sleep behaviours (5 myths); and (vi) brain function and sleep (2 myths). The experts provided feedback and rated each myth on a scale of falseness and public health importance, using a 5-point Likert scale from 1 (“not false at all”/“not significant at all”) to 5 (“extremely false”/“extremely significant”).

2.2 | Prompt and data collection methods

Like ChatGPT-4, Google Bard was tasked with verifying the truthfulness of these 20 myths and assessing their degrees of falseness and public health significance using the same 5-point scale, as described in the previous section.

More in detail, it was queried (in the technical jargon, “prompted”) about the 20 sleep-related misconceptions without prior specific training on these myths, a method known as “zero-shot learning”,

which is an ML technique where an AI-based model is required to make predictions about data that it has not seen during its training phase. As such, the model's ability to handle such queries relies only on its general understanding of language and knowledge acquired during training, without being fine-tuned on specific examples of the task at hand.

The responses generated were collected and transferred to an Excel spreadsheet for subsequent processing and analysis.

2.3 | Statistical analysis

The numbers of statements correctly classified by Google Bard as false and those eventually misclassified were computed as absolute (counts) and relative (percentage) values. The accuracy of Google Bard in classifying the 20 sleep-related statements as false was evaluated by calculating its sensitivity and positive predictive value. To compare Google Bard's performance with that of ChatGPT-4, the Fisher's Exact Test was employed to ascertain the presence of a meaningful relationship between the two categorical variables (true/false and the type of AI model), serving as an alternative to the χ^2 -test for 2×2 tables, as the cell counts were low.

The overall mean scores and standard deviations (SDs) were calculated for the 5-point Likert falseness and public health significance scales and broken down according to each domain. Google Bard's assessments of the falseness and public health significance of the 20 sleep-related myths were then compared with the evaluations of ChatGPT-4 and those provided by sleep experts. Rating distributions of the scores assigned by the two AI-based tools were compared in terms of skewness and kurtosis (computed according to Fisher's formulas). The level of agreement was gauged using inter-rater reliability analysis, specifically calculating the ICC coefficients (Liljequist et al., 2019).

All statistical analyses were conducted using the “Statistical Package for Social Sciences” (SPSS, version 28 for Windows, IBM, Armonk, NY, USA). *p*-Values of less than 0.05 were deemed statistically significant.

2.4 | Text-mining analysis

Several text-mining analyses that utilize state-of-the-art natural language processing (NLP) techniques were carried out. More in detail, based on the occurrences/frequencies of words used in the full responses to the 20 sleep-related myths, the 10 most common keywords were identified for both OpenAI ChatGPT-4 and Google Bard. Further, the two lists of word counts were compared to assess which themes or topics were more (or less) discussed and emphasized by each AI-based tool. This comparison involved looking at the frequency counts of similar words in both responses and seeing how they differed. The most commonly used words were also pictorially represented as cloud-words for both AI-enhanced chatbots.

Moreover, the three major themes for both AI-based conversational agents were identified by means of “latent semantic analysis” (LSA; Dumais, 2005), an NLP-based technique that analyses relationships between a set of documents and the terms they contain by producing a set of concepts related to the documents and terms, assuming that words that are close in meaning would appear in similar pieces of text. LSA uses singular value decomposition, a mathematical method that is leveraged to reduce the dimensionality of term-document matrices. This helps in identifying patterns and relationships in the data, and aids in understanding the context and meaning of words in large text *corpora*. In other words, although responses of both AI-based tools focused on sleep-related topics, the specific themes identified through LSA can be compared for overlaps and differences. By comparing the words that make up each theme, we can see which aspects of sleep are uniquely discussed in each AI-generated conversation and which are, instead, common to both.

Finally, overall word usage and diversity were assessed: the total number of unique words and the diversity of language used in each response were compared. This analysis includes metrics like lexical diversity (the proportion of unique words to total words). Readability was appraised by computing the Flesch Kincaid Grade Level (McClure, 1987).

All text-mining analyses were carried out using the commercial software XLSTAT (Lumivero, France).

3 | RESULTS

3.1 | Google Bard's assessment of the falseness of sleep-related false myths

Overall, the degree of falseness of sleep-related misconceptions was computed at 4.25 ± 0.70 (median 4, range 3–5; Tables 1 and S1):

TABLE 1 Google Bard's appraisals of the falseness and degree of falseness of sleep-related false myths.

Sleep-related “false myths”	Google Bard	
	True or false	On the 5-point Likert scale
Sleep duration		
“Being able to fall asleep ‘anytime, anywhere’ is a sign of a healthy sleep system”	False	5
“Many adults need only 5 or less hours of sleep for general health”	False	5
“Your brain and body can learn to function just as well with less sleep”	False	5
“Adults sleep more as they get older”	False	5
“If you can get it, more sleep is always better”	False	5
“One night of sleep deprivation will have lasting negative health consequences”	False	4
Sleep timing		
“In terms of your health, it does not matter what time of day you sleep”	False	4
Behaviours during sleep		
“Lying in bed with your eyes closed is almost as good as sleeping”	False	5
“If you have difficulty falling asleep, it is best to stay in bed and try to fall back to sleep”	False	3
“Although annoying for bed partners, loud snoring is mostly harmless”	False	4
“A sound sleeper rarely moves at night”	False	4
Daytime behaviours that relate to sleep		
“Hitting the snooze when you wake up is better than getting up when the alarm first goes off”	False	4
“If you are having difficulties sleeping, taking a nap in the afternoon is a good way to get adequate sleep”	False	4
Pre-sleep behaviours		
“Alcohol before bed will improve your sleep”	False	5
“For sleeping, it is better to have a warmer bedroom than a cooler bedroom”	False	4
“Boredom can make you sleepy even if you got adequate sleep before”	True	3
“Watching television in bed is a good way to relax before sleep”	False	4
“Exercising within 4 hours of bedtime will disturb your sleep”	False	3
Brain function and sleep		
“During sleep, the brain is not active”	False	5
“Remembering your dreams is a sign of a good night's sleep”	False	4

TABLE 2 Google Bard's appraisals of the public health significance of sleep-related false myths.

Sleep-related "false myths"	Google Bard (on the 5-point Likert scale)
Sleep duration	
"Being able to fall asleep 'anytime, anywhere' is a sign of a healthy sleep system"	3
"Many adults need only 5 or less hours of sleep for general health"	4
"Your brain and body can learn to function just as well with less sleep"	4
"Adults sleep more as they get older"	3
"If you can get it, more sleep is always better"	2
"One night of sleep deprivation will have lasting negative health consequences"	2
Sleep timing	
"In terms of your health, it does not matter what time of day you sleep"	2
Behaviours during sleep	
"Lying in bed with your eyes closed is almost as good as sleeping"	1
"If you have difficulty falling asleep, it is best to stay in bed and try to fall back to sleep"	2
"Although annoying for bed partners, loud snoring is mostly harmless"	3
"A sound sleeper rarely moves at night"	2
Daytime behaviours that relate to sleep	
"Hitting the snooze when you wake up is better than getting up when the alarm first goes off"	2
"If you are having difficulties sleeping, taking a nap in the afternoon is a good way to get adequate sleep"	2
Pre-sleep behaviours	
"Alcohol before bed will improve your sleep"	3
"For sleeping, it is better to have a warmer bedroom than a cooler bedroom"	3
"Boredom can make you sleepy even if you got adequate sleep before"	3
"Watching television in bed is a good way to relax before sleep"	2
"Exercising within 4 h of bedtime will disturb your sleep"	2
Brain function and sleep	
"During sleep, the brain is not active"	1
"Remembering your dreams is a sign of a good night's sleep"	2

Google Bard correctly identified 19 statements as false (95.0%), not differing from ChatGPT-4 (85.0%), as evidenced by the statistical test (Fisher's exact test $p = 0.615$).

Of note, although the accuracy rates were comparable, differences in the rating distributions of the two AI-based tools could be noted. More in detail, the skewness and kurtosis for the ratings given by Bard are as follows: -0.42 and -0.83 , indicating a moderately negatively skewed distribution, similar to the normal distribution in terms of its tails' heaviness, with slightly lower ratings compared with higher ones, slightly lighter tails than the normal distribution, and fewer extreme values.

While both distributions were relatively flat compared with the normal distribution, with respect to ChatGPT-4 (skewness and kurtosis of -0.23 and -0.74 , respectively), Google Bard's ratings showed a slightly more pronounced skewness, with more ratings on the higher end, and kurtosis, with a slightly wider, more flat-topped

distribution than that of ChatGPT-4, indicating a bit more variability in ratings.

3.2 | Google Bard's assessment of the public health significance of sleep-related false myths

Overall, the degree of public health significance of the 20 sleep-related false myths yielded a value of 2.4 ± 0.80 (median 2, range 1–4; Table 2).

Skewness and kurtosis yielded values of 0.36 and -0.07 , respectively, which differed from those of the distribution of ChatGPT-4's ratings (0.40 and -0.77). This indicates that, while both distributions are positively skewed, the ChatGPT-4 distribution is slightly more skewed towards higher values. Moreover, the Google Bard distribution is closer to a normal distribution, whereas

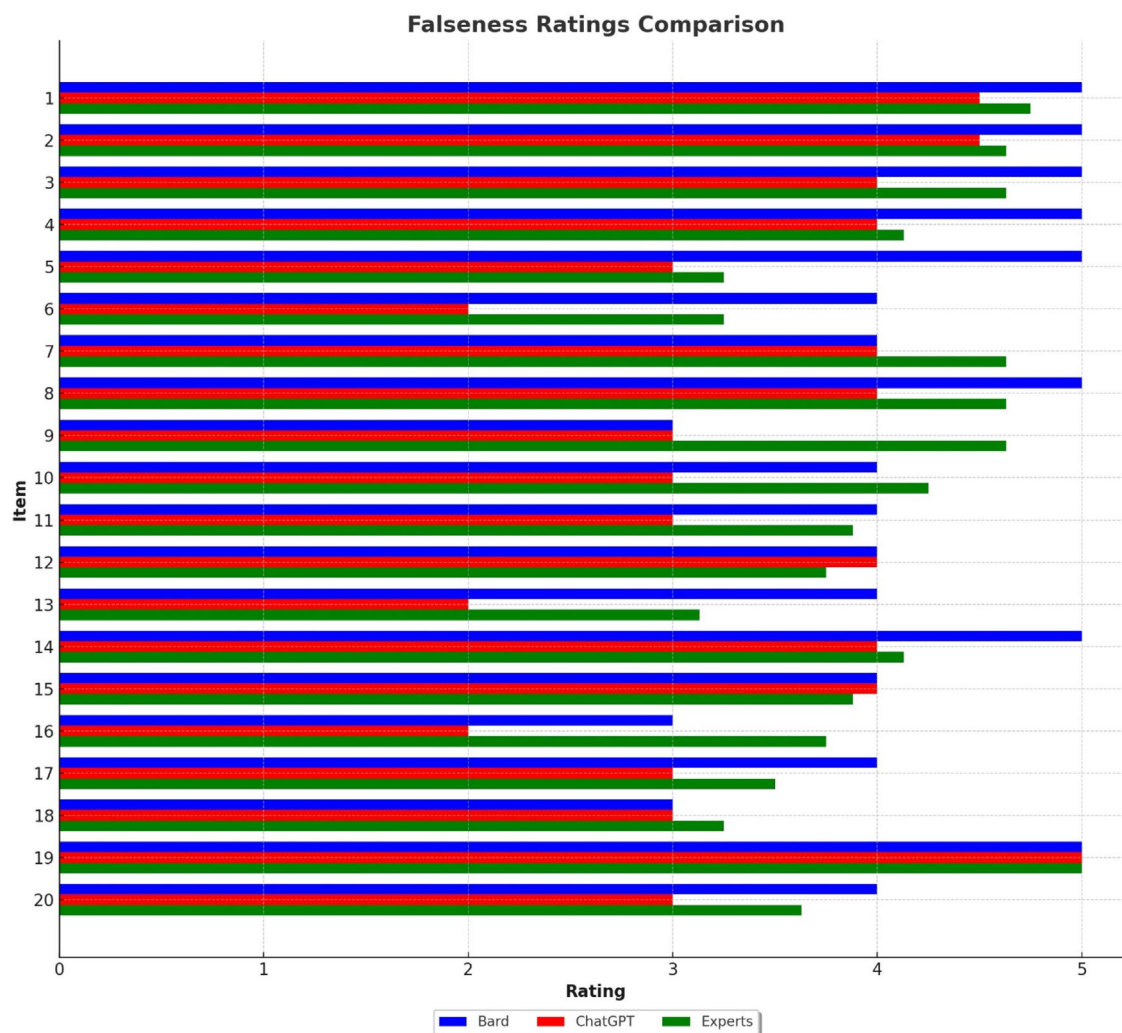


FIGURE 1 Item-by-item comparison of ratings provided by sleep experts and artificial intelligence (AI)-based tools (OpenAI ChatGPT-4 and Google Bard) on the falseness of 20 sleep expert-identified sleep-related myths. The full content of the items is reported in Table 1.

the ChatGPT-4 distribution has lighter tails, indicating fewer extreme ratings.

3.3 | Reliability and inter-rater agreement among AI-based tools and sleep experts

From a qualitative perspective, Google Bard tends to give higher ratings compared with ChatGPT-4 and experts in most domains. This is especially noticeable in the “Sleep duration” and “Behaviors during sleep” categories. ChatGPT-4’s ratings are generally more aligned with the experts, although there are differences in some areas, particularly in “Daytime behaviors that relate to sleep” and some aspects of “Pre-sleep behaviours”.

From a quantitative standpoint, the degree of inter-rater agreement on the falseness of the sleep-related statements between the two AI-based tools was computed at 0.78 (95% confidence interval [CI]: 0.45–0.91; $p = 0.002$). Concerning the consistency between

Google Bard’s and experts’ ratings, the ICC analysis yielded a value of 0.58 (95% CI: -0.06 to 0.83 ; $p = 0.065$), as pictorially shown in Figure 1.

Concerning the public health significance of the 20 sleep-related false myths, in several domains like “Sleep duration” and “Behaviors during sleep”, the ratings differ noticeably among the three groups. Google Bard tends to give lower ratings in many cases, particularly in “Behaviours during sleep” and “Brain function and sleep”. There are instances where ChatGPT-4’s ratings align more closely with the experts (e.g. “Daytime behaviors that relate to sleep” and “Pre-sleep behaviours”), and other instances where they diverge (e.g. “Brain function and sleep”). Of note, Google Bard’s ratings are quite variable: in some domains like “Sleep duration”, Google Bard’s ratings are closer to the experts, while in others like “Brain function and sleep”, Google Bard’s assigned scores are lower.

Quantitatively, the agreement on the public health significance of the 20 sleep-related “false myths” between the two chatbots was computed at 0.47 (95% CI: -0.34 to 0.79 ; $p = 0.177$), which was not

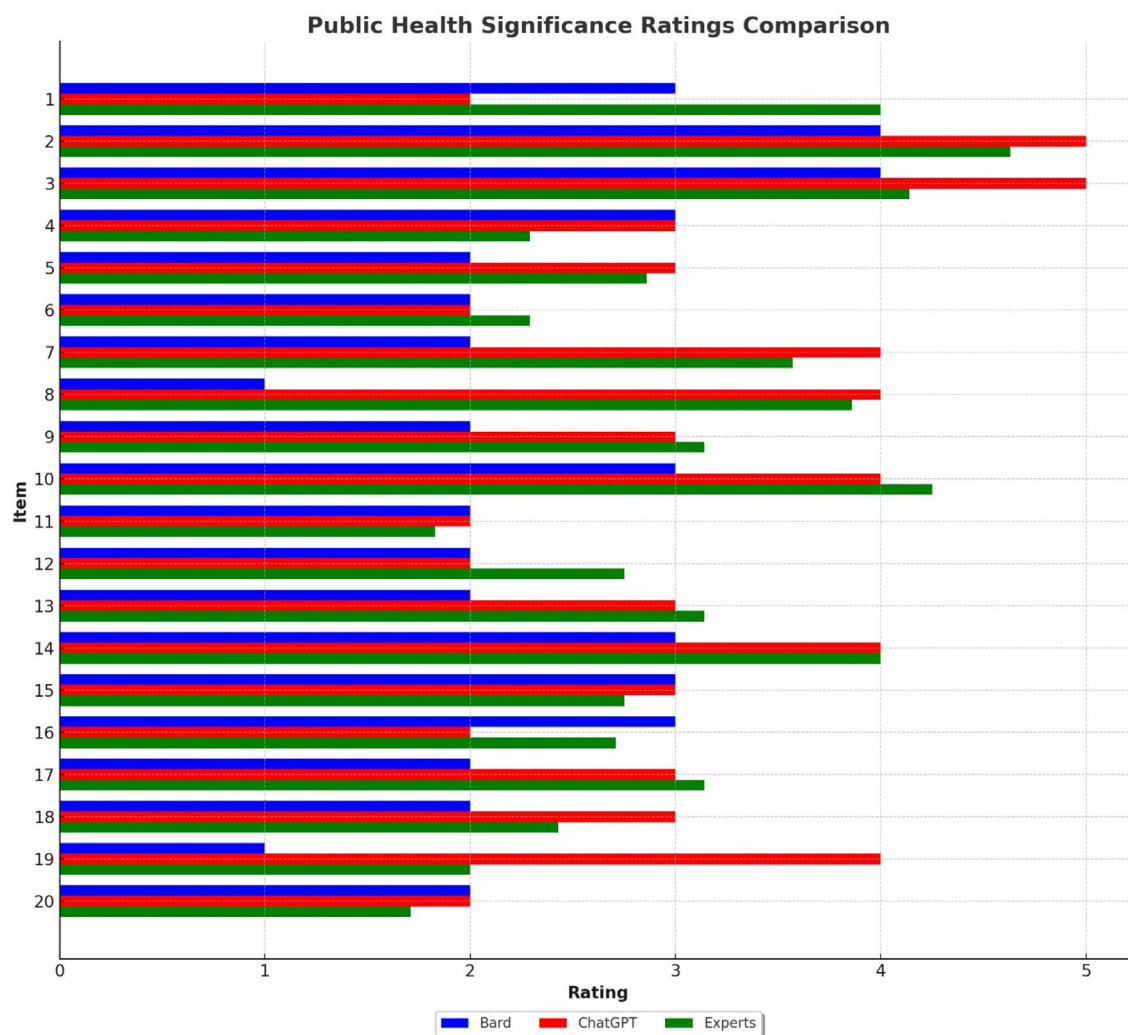


FIGURE 2 Item-by-item comparison of ratings provided by sleep experts and artificial intelligence (AI)-based tools (OpenAI ChatGPT-4 and Google Bard) on the public health significance of 20 sleep expert-identified sleep-related myths. The full content of the items is reported in Table 2.

statistically significant. The ICC coefficient of the agreement between Google Bard and sleep experts was 0.69 (95% CI: 0.22–0.88; $p = 0.014$; Figure 2).

3.4 | Cloud-word analysis and NLP-based thematic modelling

Overall, the cloud-word analysis of the responses provided by OpenAI ChatGPT-4 and Google Bard to the 20 sleep-related myths revealed the following themes: (i) sleep quality as a major concern, with many myths related to how to get a good night's sleep; (ii) sleep hygiene as another important theme, with myths related to what to do before and after bed; and (iii) sleep disorders as another topic of concern, with myths related to the symptoms and causes of sleep problems.

Specifically concerning Google Bard, the following are the 10 most common words and phrases that appeared in the cloud-word analysis

(Figure 3): (i) sleep; (ii) fall(ing) asleep; (iii) bed; (iv) help; (v) night; (vi) brain; (vii) feel(ings); (viii) avoid; (ix) tips; and (x) alcohol. The NLP analysis of the responses to the 20 sleep myths revealed the following themes: (i) "Practical Sleep Strategies"; (ii) "Sleep Challenges and Solutions"; and (iii) "Lifestyle and Environmental Impact on Sleep". The first theme is centred on providing practical advice and strategies to improve sleep quality. The presence of words like "sleep", "bed", "fall (ing) asleep", "help" and "tips" suggests a focus on actionable steps and habits that individuals can adopt to enhance their sleep. The presence of words like "avoid" and "brain" indicates discussions around avoidance of habits and practices that might disrupt sleep and the neurological aspects of sleep, respectively. This theme likely covers a range of sub-topics, from setting up a conducive sleep environment (e.g. comfortable bed) to understanding how certain behaviours or routines (e.g. avoiding caffeine or screen time before bed) can impact one's ability to fall and stay asleep. The second topic can be interpreted as focusing on different aspects of sleep challenges and their

approaches and specific emphases differ, with ChatGPT-4 possibly leaning more towards an informational perspective and Google Bard towards practical advice.

3.6 | Readability analysis

The Flesch–Kincaid Grade Level analysis indicated differences in the readability of the responses provided by Google Bard and OpenAI ChatGPT-4, suggesting that the two conversations were comparable in complexity to those typically found in 8th-grade and 12th-grade (or senior year of high school) level material, respectively.

4 | DISCUSSION

4.1 | Study findings

The present study compared the assessments of sleep-related “false myths” provided by two major AI-based tools, Google Bard and ChatGPT-4, with a focus on their ratings of the falseness and public health significance of these myths, as well as their inter-rater agreement with sleep experts. The ICCs for falseness seem to suggest only a moderately strong relationship between AI and experts' ratings, indicating that AI-based tools are not yet highly efficacious in detecting sleep-related misinformation. Furthermore, there is a noticeable difference in the distribution of ratings given by Google Bard and ChatGPT-4. Google Bard's ratings show more variability and are slightly more skewed towards higher values in terms of falseness and closer to a normal distribution in terms of public health significance. Moreover, the level of agreement between the two AI tools and between AI and experts varies depending on the aspect being rated (falseness versus public health significance). This suggests that, while AI tools are in substantial agreement, their assessments may still differ from human expert opinions in evaluating the falseness and public health significance of sleep-related myths. Overall, the present analyses indicate that the assessments of false statements on sleep from AI tools like Google Bard and ChatGPT-4 can vary both in terms of falseness and public health implications, underscoring the importance of combining AI assessments with expert human judgement in health-related contexts.

4.2 | Differences and similarities between ChatGPT-4 and Google Bard

The AI-based tools (ChatGPT-4 and Google Bard) can vary significantly in their assessments. Such variability might stem from differences in their programming, transformer architecture, data sources or algorithms used for analysis. Also, the differences between AI and experts' ratings suggest that while AI tools are valuable for providing quick assessments, they may not always align with expert opinions, which are likely based on more comprehensive, nuanced

understandings of the topics. Further, according to the statistical analyses (reliability and inter-rater agreement analysis), the ratings across different domains suggest that the accuracy or alignment of AI tools with expert opinions can be highly context-dependent. This indicates the importance of considering AI tool ratings as one of many inputs, especially in fields like sleep science. Overall, these data underscore the need for careful interpretation of AI assessments, especially in complex, evolving fields, and highlight the value of human expertise in providing balanced and contextually informed judgements.

The text-mining analysis brought forth interesting insights into the thematic differences between the two AI tools. Both Google Bard and ChatGPT-4 discussed themes like sleep quality, hygiene and disorders. Google Bard, however, leaned more towards practical advice, while ChatGPT-4 focused more on the theoretical (mainly, physiological) aspects of sleep. The analysis of word usage highlighted differences in emphasis and frequency of certain keywords, underlining the distinct approaches of Google Bard and ChatGPT-4 in discussing sleep-related topics. Lastly, the readability analysis compared the complexity of responses from both AI tools, suggesting that, while ChatGPT-4's responses were at a higher grade level, Bard's responses were comparatively simpler and more accessible to a broader audience.

In essence, this comparative study shines a light on the effectiveness and uniqueness of the approaches of AI tools like ChatGPT-4 and Google Bard in disseminating sleep-related information and debunking myths, showcasing the potential of AI in public health education, particularly in the realm of sleep health, while also drawing attention to the necessity of ensuring accuracy and dependability in the information these tools provide.

4.3 | Implications for future AI development in sleep health education

The findings of this study have significant practical implications for the future development of AI in the field of sleep health education: they suggest a growing role for AI in providing accessible, diverse and reliable health information, but also emphasize the need for continuous improvement in AI algorithms to ensure the accuracy and relevance of the information provided (Monteith et al., 2024).

The study encourages further collaborative efforts between AI developers, sleep health professionals and educators to leverage the strengths of AI tools like ChatGPT-4 and Google Bard. Such collaborations can lead to more nuanced and comprehensive sleep health education strategies, particularly in online and digital formats.

5 | CONCLUSIONS

The current study demonstrated the transformative potential of AI tools in addressing and correcting common myths and misconceptions about sleep and, broadly, disseminating crucial health-related information. Through their distinct approaches, ChatGPT-4 and Google Bard

effectively engage in public health education, particularly in the domain of sleep health. The present study also highlights the unique methodologies employed by ChatGPT-4 and Google Bard: while the former tends to focus more on theoretical aspects and detailed explanations of sleep health, the latter leans towards providing more practical advice. This diversity in approaches enhances the spectrum of information available to the public.

Finally, the comparative analysis underlines the importance of accuracy and reliability in the information provided by AI tools. Given the impact of misinformation on public health, especially in sensitive areas like sleep health, the study stresses the need for these AI-based platforms to maintain high standards of evidence-based, factual correctness.

AUTHOR CONTRIBUTIONS

Sergio Garbarino: Conceptualization; investigation; writing – original draft; methodology; validation; visualization; writing – review and editing; supervision; funding acquisition. **Nicola Luigi Bragazzi:** Conceptualization; investigation; funding acquisition; writing – original draft; methodology; validation; visualization; writing – review and editing; software; formal analysis; project administration; resources; supervision; data curation.

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CONFLICT OF INTEREST STATEMENT

None to declare.

DATA AVAILABILITY STATEMENT

The data that supports the findings of this study are available in the supplementary material of this article.

ORCID

Sergio Garbarino  <https://orcid.org/0000-0002-8508-552X>

Nicola Luigi Bragazzi  <https://orcid.org/0000-0001-8409-868X>

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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